

KEVIN LI

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science | GPA: 3.70

Orlando, FL

Aug 2029

EXPERIENCE

Software Engineer Lead

Jul. 2026 – Present

Knights Design Interactive

Orlando, FL

- Led a team of **6** developers through two-week **Agile** sprints and weekly standups, running sprint planning and unblocking devs on technical issues to ship a member portal in one semester and grow active members **35%**.
- Architected the member portal for **300+** members, building custom event creation and sign-up flows on a **PostgreSQL** backend with type-safe **Drizzle ORM** schemas in **TypeScript** that cut auth and schema bugs **50%**.
- Introduced a PR template, review checklist, and reviewer rotation, cutting revision cycles from **4** rounds to **2** and raising merged PRs from **2** to **5** per week to keep weekly releases on schedule.

Software Engineer

Mar. 2026 – Present

UCF SASE

Orlando, FL

- Reduced sign-up time **20%** by implementing unified **OAuth2** single sign-on in **TypeScript** for Discord and Google alongside email/password auth on **Supabase**, standardizing authentication for **700+** chapter members.
- Cut member check-in time **75%** by replacing a Google Form with QR-code sign-in on a connected database, and built **Node.js Discord** bots that auto-post event reminders from the live calendar, raising attendance visibility **20%**.

Software Engineer Intern

May 2026 – Jul. 2026

UCF Institute for Simulation & Training

Orlando, FL

- Built a dual-perspective (macro + first-person) rendering system with real-time switching across **10+** radiation sources, shipped as an open-source prototype for ORAU that saves **\$2,500** per setup over physical lab equipment.
- Engineered a real-time VR radiation simulator in **Unity (C#)** accurately rendering alpha, beta, and gamma particle trail physics, achieving stable **90 FPS** on Meta Quest 3.

PROJECTS

Rescuebot | *Python, C++, Raspberry Pi 5, ESP32-S2, YOLO11, Computer Vision*

- Built an autonomous search-and-rescue robot that maps rooms with LiDAR, finds people, and reports who needs help, writing the **Python** vision pipeline that runs **YOLO11n** on the camera's own chip to detect **99%+** of people in real time.
- Integrated the hardware, wiring a **Raspberry Pi 5** to **2** cameras, LiDAR, and an **ESP32-S2** running **C++** mecanum-wheel control for **4** motors via a motor shield, plus gyroscope, temperature, and PIR sensors on isolated Pi/motor power.

Mimir | *Next.js, React, TypeScript, Python, FastAPI, LiveKit, Gemini, Docker*

- **Won 2nd Overall at SASEHacks out of 167 participants** by building a voice-assisted AI whiteboard where students solve math on an infinite canvas while a live **Gemini** tutor talks them through each step and annotates their work.
- Streamed canvas strokes and pasted images to the tutor over **LiveKit** with in-browser **Tesseract.js** OCR, and built tools that convert math equations into interactive graphs, plus physics animations tested on **200+** cases.

CrisisNet | *Next.js, React, TypeScript, Tailwind CSS, Google ADK*

- **Won Best Use of .tech Domain at HackUSF 2026** by architecting an AI-powered multi-agent disaster response platform for Tampa Bay flood resilience in a **4-person** team, shipping bilingual English/Spanish alerts.
- Built a **Next.js/TypeScript** dashboard with live maps plotting **48** shelters and historical flood charts, integrated with a **Google ADK** backend orchestrating **5** AI agents that score incidents P1–P3 and send personalized alerts via **Mailgun**.

Fleurish | *Supabase, PostgreSQL, PostGIS, Deno, TypeScript, React Native*

- Built a gamified social plant-identification app, implementing its core features (photo capture, personal garden, leaderboard, and social feed) across **8** Deno edge functions and a **9-table Supabase** PostgreSQL schema.
- Engineered a two-stage AI pipeline where **Plant.id v3** identifies species and **Gemini 2.5 Flash** diagnoses disease, plus a **PostGIS** proximity search returning distance-sorted sightings in a **single** round-trip.

Vigil | *Next.js, React, TypeScript, WebSockets, FastAPI, SQLite*

- Built the frontend for an open-source, self-hosted **SIEM** developer tool delivering real-time monitoring across **4** live views (events, alerts, health, settings) with a **6-developer Agile** team tracked in **Jira**.
- Delivered sub-second event streaming by connecting a **FastAPI + SQLite** backend to the **Next.js** client over **WebSockets**, ingesting logs through **120+** Grok patterns with **OpenAI**-generated fallbacks.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, C#, SQL, HTML, CSS

Frameworks / Libraries: React, Next.js, React Native, Node.js, Tailwind CSS, FastAPI, Drizzle ORM, PyQt6, Unity, Three.js

Databases: PostgreSQL, PostGIS, SQLite, Supabase

Developer Tools: Git, GitHub, GitHub Actions, Docker, Vercel, Deno, WebSockets, REST APIs, OAuth2, Jira